To Be Ahead

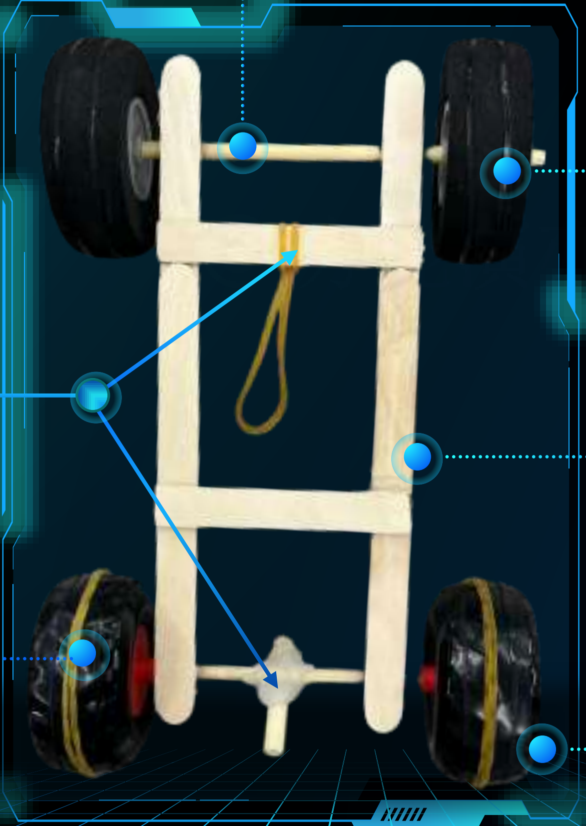
The Less The Better

Self-powered Trolley Leader

Presenter:
Mengfei Zhang Yunyi Zhang
Jingyu Chen Weizhen Xu

Structural Framework

Weight : 60g



Back-wheel drive
Rubber band drived

Material Selection & Sources

Reasons for selection

"#" bracket

Robustness
Light Degradable

Crossbar & Power Driving System

Robustness
Light Degradable

Power Driving System & Tyre Pattern

High Elasticity & Friction
Recyclable

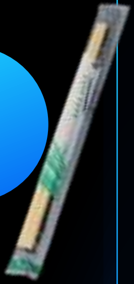
Front and Back Wheels

Light Stable Recyclable

Materials

Ice-cream Stick

Bamboo Chopsticks



Rubber Bands

Aircraft model wheels

Source of material

Leftovers from Zhang Yunyi's high school crafts.

From the restaurants in Hoi Tin Court & Sing Tin Court.

Bought at the QM shop.

Removed from Two model aeroplane.

Material Selection & Sources

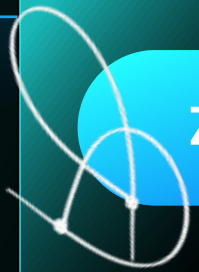
Reasons for selection

Connecting the Wheel to Bracket

Light Solid Degradable

Tyre Pattern

Light High friction Recyclable



Zip Ties

Little Black Leather Band

Source of material

Left over from the club's model assembly

Small leather bands to tie your hair.

Weight

60g

Sustainable

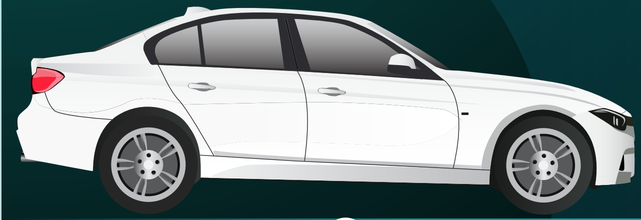
Cost

QM 0.6

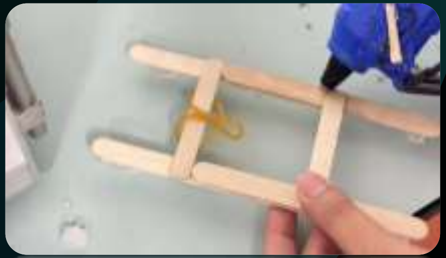
The Less The Better

Connection Procedure

To Be Ahead



“#” bracket



Simple & Robust

Rubber-band Binding

Simple & User-friendly

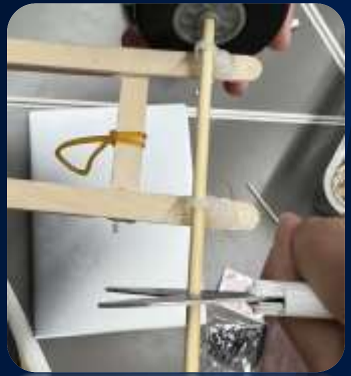
Increase tyre tread pattern

Preventing tire slippage

Ties are looped & secured to the bracket

Simple & Robust

Crossbar to car tyres



Robust

User-friendly

Assembling

To Be Ahead

Power Driving System

Front Bracket & Rubber Band & Rear Crossbar

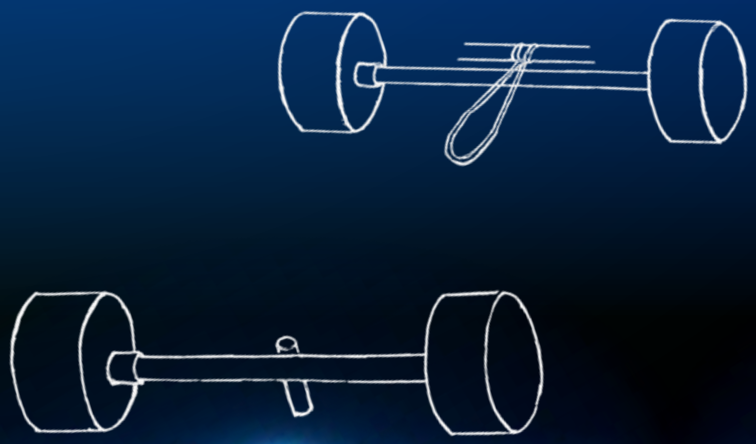
Back-wheel drive

Driving

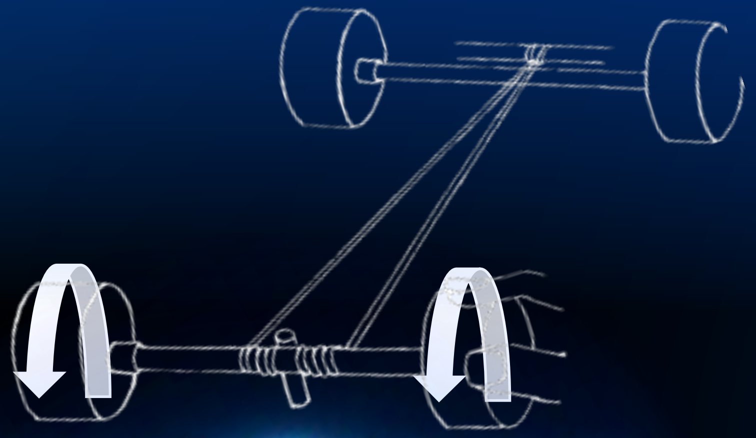
Annual Winding	Rubber Band Stretching	Releasing & Moving

Energy

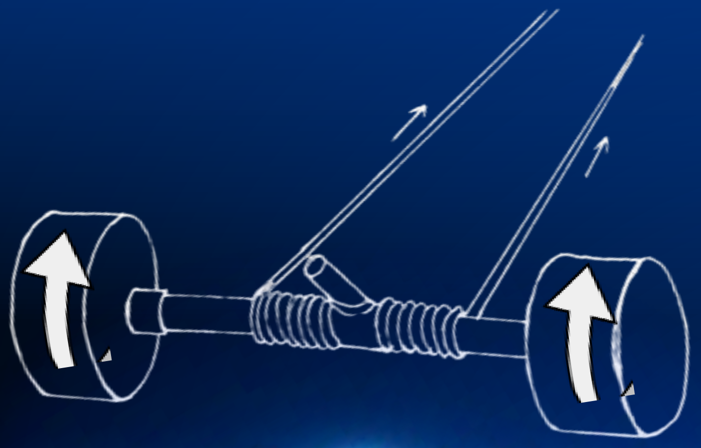
Human Mechanical Energy	Elastic Potential Energy of Rubber Bands	Car Kinetic Energy



Power Driving System



Loading



Releasing

To Be Ahead

Calculation of energy storage

1. Data preparation

Table I. Parameters of the cart

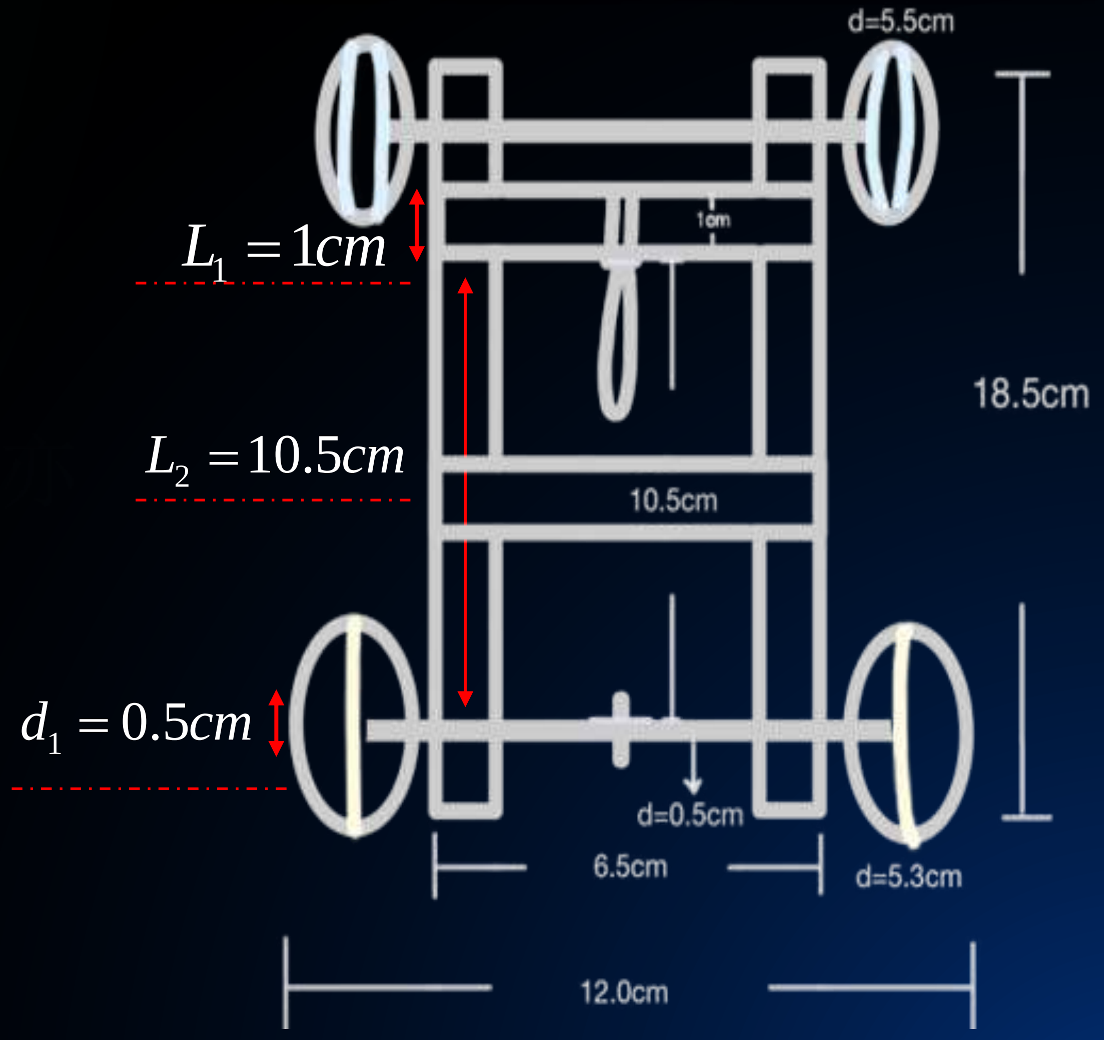
Original length of rubber band	L_0=6cm
Beam length	L_1=1cm
Length from the fixing point of the leather band to the crossbar	L_2=10.5cm
Crossbar	Calibr: d_1=0.5cm Round: $L_3=\pi d_2$

$$L = 2L_1 + L_2 + nL_3$$

(nNumber of rubber band wraps)

$$\Delta L = L - L_0$$

$$\Delta L = 2L_1 + L_2 + nL_3 - L_0$$



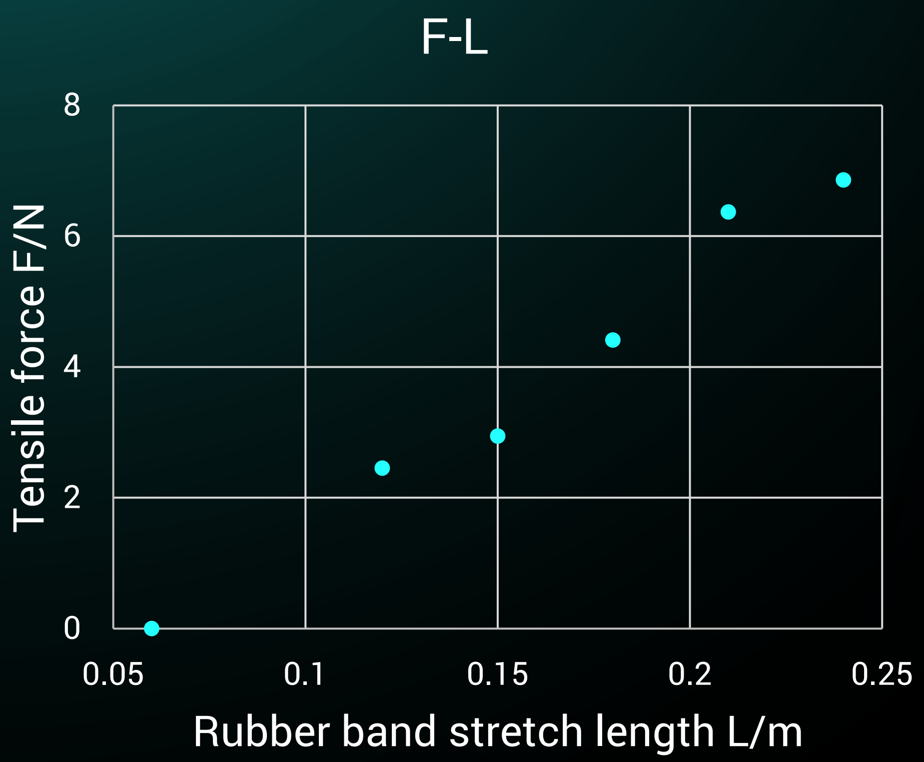
Cart parameter diagram

To Be Ahead

Calculation of energy storage

1. Data preparation $$\Delta L = 2L_1 + L_2 + nL_3 - L_0$$

2. Calculation of the coefficient of strength k



Rubber Band Tension-Length Chart

Large variation in k values of rubber bands at different nodes

Taking the values at the endpoints k

Table II. Coefficients of strength of rubber bands

Rubber band stretch length/m	Coefficient of strength k (N/m)
0.12	40.83
0.15	32.67
0.18	36.75
0.21	42.47
0.24	38.11

To Be Ahead

Calculation of energy storage

1. Data preparation

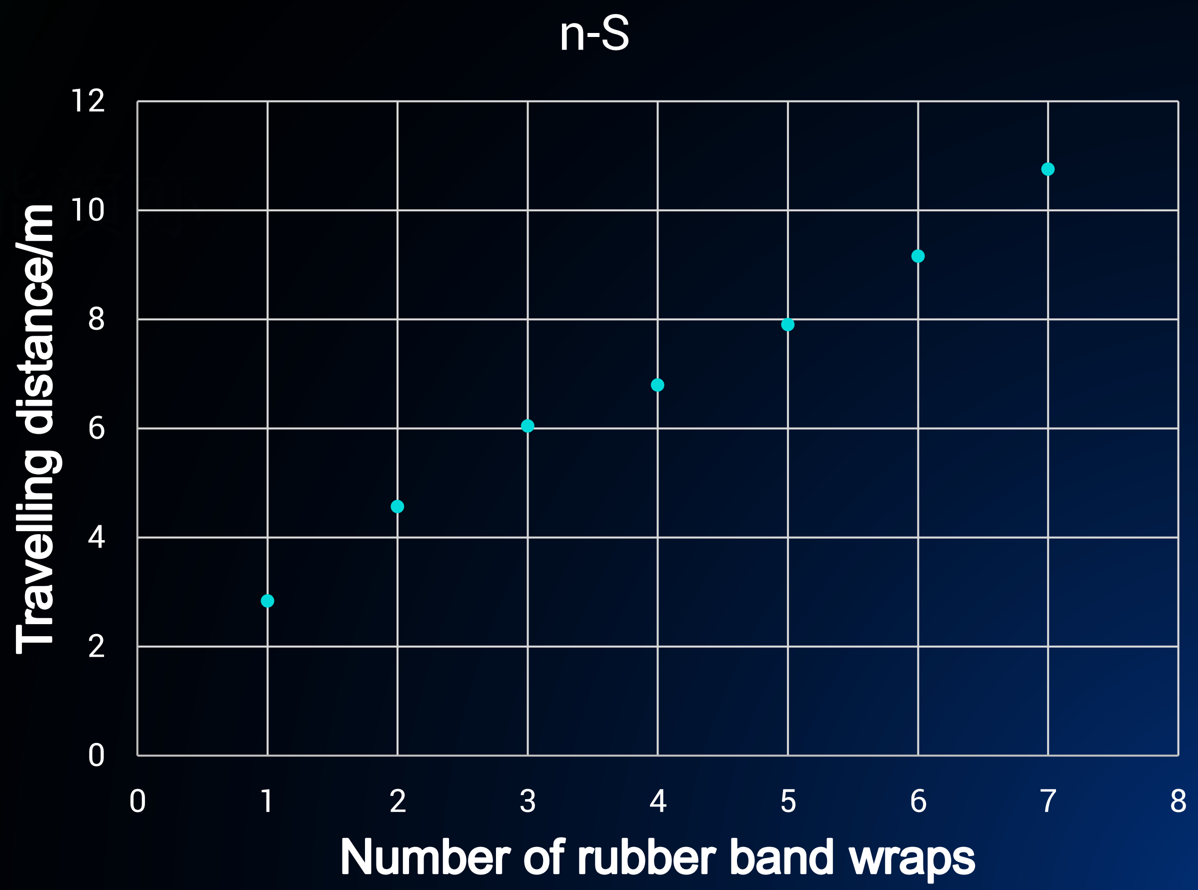
2. Calculation of the coefficient of strength k

3. Energy storage calculations & Pre-testing

$$E_{\text{Stored}} = \frac{1}{2}k(\Delta L)^2$$

Table III. Test data

Number of rubber band wraps n	Travelling distance S/m	Travelling time/s
1	2.84	4.81
2	4.57	7.41
3	6.05	8.13
4	6.80	8.36
5	7.91	8.81
6	9.16	9.13
7	10.76	9.76



Calculation of energy storage

To Be Ahead

4. Energy conversion ratio

Target distance: $6.7 \pm 0.35m(5\%)$

When n = 4, the distance travelled is 6.8m

Number of turns	4/rads
Travelling time	8.36/s
Travelling distance	6.80/m

Data preparation $n = 4$

$$\Delta L = 2L_1 + L_2 + nL_3 - L_0 \qquad \Delta L = 0.12783cm$$

$$L = 2L_1 + L_2 + nL_3 \qquad L \approx 18cm$$

Energy storage calculations

$$L \approx 18cm \qquad k = 36.75N / m$$

$$E_{\text{Stored}} = \frac{1}{2}k(\Delta L)^2 \qquad E_{\text{Stored}} = 0.3003J$$

Dynamics calculation of kinetic energy

$$S = 0.5at^2 \qquad E_d = 0.0806J$$

$$v_0 = at \qquad \eta = \frac{E_d}{E_{Stored}}$$

$$E_d = 0.5mv^2$$

Energy conversion ratio

26.8%

To Be Ahead

THANKS!

Presenter： M18
PPT： Mengfei Zhang